Cpts 327 Assignment 5

Instructions

- **Format Requirement:** Please round (nor floor or ceil) all numerical answers to **two decimal places** (Please use fractions in intermediate steps to avoid precision loss and only round at the very end), e.g., round the result 4.555 to 4.56, 4.444 to 4.44. If your calculated answer has fewer than two decimal places (e.g., 0.1), you must pad it with a trailing zero (e.g., enter **0.10**) to ensure the grading system accepts it. Please express all answers **involving modular arithmetic (e.g., $\pmod{n}$) as the smallest integer in the range $[0, n-1]$.**

Problem 1: (8 Points) Consider the multiplicative group $G = (\mathbb{Z}_{19}^*, \cdot)$.

(a) True or False: The group G is a cyclic group.

(b) What is the order of the group $|G|$?

(c) True or False: The element $g = 2$ is a generator of G.

(d) Calculate $2^{100} \pmod{19}$.

Problem 2: (8 Points) Consider the multiplicative group $G = (\mathbb{Z}_{21}^*, \cdot)$.

(a) True or False: The group G is a cyclic group.

(b) What is the order of the group $|G|$?

(c) True or False: The element $g = 2$ is a generator of G.

(d) Calculate $4^{100} \pmod{21}$.

Problem 3: (8 Points) Consider the multiplicative group $G = (\mathbb{Z}_{33}^*, \cdot)$.

(a) True or False: The group G is a cyclic group.

(b) What is the order of the group $|G|$?

(c) True or False: The element $g = 2$ is a generator of G.

(d) Calculate $2^{122} \pmod{33}$.

Problem 4: (16 Points) Let $p = 23$ and consider the subgroup G of quadratic residues in $\mathbb{Z}_{23}^*$.

(a) True or False: This subgroup G is a cyclic group.

(b) What is the order of this subgroup $|G|$?

(c) True or False: The element $g = 2$ is a generator of this subgroup G.

(d) Calculate $18^{122} \pmod{23}$.

Problem 5: (16 Points) Let $p = 101$ and consider the multiplicative group $G = (\mathbb{Z}_{101}^*, \cdot)$. We want to find the smallest positive integer x such that:

$$2^x \equiv 10 \pmod{101}$$

(a) True or False: The order of the element $g = 2$ in $\mathbb{Z}_{101}^*$ is 100.

(b) Calculate $2^{50} \pmod{101}$.

(c) Calculate $2^{20} \pmod{101}$.

(d) What is the value of x?

Problem 6: (12 Points) Alice and Bob agree to use the textbook Diffie-Hellman key exchange protocol with the public parameters $p = 127$ (modulus) and $g = 2$ (generator).

(a) Alice chooses her private key $a = 40$. Calculate her public key $A = g^a \pmod{p}$.

(b) Bob chooses his private key $b = 15$. Calculate his public key $B = g^b \pmod{p}$.

(c) Alice receives B and computes the shared secret $K = B^a \pmod{p}$. Calculate K.

Problem 7: (12 Points) Alice and Bob want to modify the texbook Diffie-Hellman protocol in Problem 6 with a large prime $q = 2^{255} - 19$ as the modulus. The modulus looks large and secure. And Alice wants to use a lucky number $h = 8$ as the base instead of $g = 2$.

(a) Alice chooses a new private key $a = 85$. Calculate the new public key $A = h^a \pmod{q}$.

(b) Bob's chooses a new private key $b < a$ and calculate the new public key $B = h^b \pmod{q}$. The shared secret key is $K = B^a \pmod{q}$. If an attacker obtains K, and tries to recover the private key b by brute-force searching around 2^{255} possible answers, and its computer can check 2^{80} keys per second, approximately how many years would it take? Assume 1 year $\approx 2^{25}$ seconds, the total time is 2^t years, calculate t.

(c) A malicious attacker Eve eavesdrops the shared secret key $K = 361$. Help Eve to recover b.

Problem 8: (12 Points) Alice uses the ElGamal encryption scheme with the public parameters $p = 101$ (modulus) and $g = 2$ (generator). Her public key is $h = 89 \pmod{101}$.

(a) Calculate Alice's private key x, which is the smallest integer satisfying $g^x \equiv h \pmod{101}$.

(b) Bob wants to send a message $m = 5$ to Alice. He chooses a random key $k = 50$. Calculate the ciphertext (c_1, c_2), where:

$$c_1 = g^k \pmod{101}, \quad c_2 = m \cdot h^k \pmod{101}$$

(c) The decryption process contains $s = c_1^x \pmod{101}$ and $m = c_2 \cdot s^{-1} \pmod{101}$. Calculate s.

Problem 9: (4 Points) Alice and Bob decide to improve the efficiency of the ElGamal encryption by reusing the same random ephemeral key k for all messages $m_1, m_2, \ldots, m_n$ sent during a single session. For a public key $h = g^x \pmod{p}$, the ciphertext sequence is:

$$C_i = (c_{1,i}, c_{2,i}) = (g^k \pmod{p}, m_i \cdot h^k \pmod{p})$$

where k is chosen once at the beginning of the session and remains secret.

Q: True or False: This variant remains semantically secure (IND-CPA).

Problem 10: (4 Points) Consider a modified ElGamal scheme where the message m is not a group element but a bit-string. Let $H : \mathbb{Z}_p^* \to \{0,1\}^\ell$ be a cryptographic hash function. The encryption of $m \in \{0,1\}^\ell$ is:

$$C = (c_1, c_2) = (g^k \pmod p, m \oplus H(h^k \pmod p))$$

where $\oplus$ denotes the bitwise XOR operation.

Q: True or False: This variant remains semantically secure (IND-CPA).